Answer Key - Mathematics - Graduate Level Assessment 12 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 900; B) 441; C) 300; D) 484; E) 529
   Note: The least perfect square with three different prime factors is 900 because it can be expressed as 30^2 (where 30 is the product of the three distinct primes 2, 3, and 5), ensuring that each prime factor contributes to the square while maintaining the least value.

2. **Answer:** B
   Options: A) The column vectors of M form a basis for R^5.; B) For any two distinct column vectors u and v of M, the set {u, v} is linearly independent.; C) The trace of M is nonzero.; D) The eigenvalues of M are all nonzero.; E) The homogeneous system Mx = 0 has only the trivial solution.
   Note: The condition stating that the set {u, v} is linearly independent for any two distinct column vectors u and v of M is not equivalent to the other four conditions because it refers to the linear independence of specific column pairs rather than the overall rank or nullity properties that characterize the other conditions.

3. **Answer:** D
   Options: A) (0, 0); B) (6, 0); C) (9, 16); D) (0, 9); E) (3, 6)
   Note: The correct answer is (0, 9) because for the quadratic inequality $x^2 - 6x + c < 0$ to have real solutions, the discriminant $D = (-6)^2 - 4(1)(c) = 36 - 4c$ must be positive, leading to the condition $c < 9$, while c must also be positive, hence the interval for c is (0, 9).

4. **Answer:** D
   Options: A) 2 x + 2 x and 2 x^2; B) x+x+x+x and x^4; C) 3 x + 5 x and 15 x; D) 4(2 x - 6) and 8 x - 24; E) 3 x + 3 x + 3 x and 3 x^3
   Note: The expressions 4(2 x - 6) and 8 x - 24 are equivalent because, when the distributive property is applied to the first expression, it simplifies to 8 x - 24, demonstrating that they represent the same linear function.

5. **Answer:** D
   Options: A) $\frac{17}{31}$; B) $\frac{33}{66}$; C) $\frac{33}{100}$; D) $\frac{17}{50}$; E) $\frac{1}{6}$
   Note: The correct answer is $\frac{17}{50}$ because there are 50 integers in the set $\{1, 2, \ldots, 100\}$ that are divisible by 2, and among them, 17 are not divisible by 3, leading to a probability of $\frac{17}{100}$ which simplifies to $\frac{17}{50}$.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because when you add the fractions 4/7 and 2/7, you combine the numerators while keeping the denominator the same, resulting in (4 + 2)/7 = 6/7.

7. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the conversion factor between baps and daps indicates that 42 baps equals 40 daps, confirming the relationship between the two units.

8. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the area between the curves f(x) and g(x) from x = 1 to x = 3 can be accurately calculated by integrating the difference of the functions over that interval, yielding a result of approximately 2.987.

9. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because 26.1 mm is equal to 2.61 dm, not 0.0261 dm, as there are 100 mm in a cm and 10 cm in a dm, making 1 dm equal to 100 mm.

10. **Answer:** True
    Options: A) True; B) False
    Note: The value of y is 3 because it maintains the proportionality of corresponding sides of the similar polygons, confirming that the similarity condition is satisfied.

Long Form Response

11. **Answer:** To deal with the $|x|$ term, we take cases on the sign of x: If $x \geq 0$, then we have $y^2 + 2xy + 40x = 400$. Isolating x, we have $x(2y + 40) = 400 - y^2$, which we can factor as
    $$2x(y + 20) = (20 - y)(y + 20).$$
    Therefore, either $y = -20$, or $2x = 20 - y$, which is equivalent to $y = 20 - 2x$. If $x < 0$, then we have $y^2 + 2xy - 40x = 400$. Again isolating x, we have $x(2y - 40) = 400 - y^2$, which we can factor as
    $$2x(y - 20) = (20 - y)(y + 20).$$
    Therefore, either $y = 20$, or $2x = -y - 20$, which is equivalent to $y = -20 - 2x$. Putting these four lines together, we find that the bounded region is a parallelogram with vertices at $(0, \pm 20)$, $(20, -20)$, and $(-20, 20)$, as shown below: [asy]size(6 cm);real f(real x) {return 20; } draw(graph(f, -25, 0)); real g(real x) { return -20; } draw(graph(g, 0, 25)); real h(real x){return 20-2*x;} draw(graph(h, 0,25));

real i(real x){return -20-2*x;} draw(graph(i, -25,0)); draw((0,-32)–(0,32),EndArrow); draw((-26,0)–(26,0),EndArrow); label("x",(26,0),S); label("y",(0,32),E); dot((0,20)–(0,-20)–(20,-20)–(-20,20));[/asy] The height of the parallelogram is 40 and the base is 20, so the area of the parallelogram is $40 \cdot 20 = 800$.

Note: correct because it systematically considers the absolute value function by breaking it into two cases based on the sign of x, allowing for the proper isolation and manipulation of variables to derive the equations that describe the bounded regions in the plane.

12. **Answer:** To derive the function h(x) = (x$^{\{}-1/2\}+2x)(7-x^{\{}-1\}), wewillapplytheproductruleandthe$
$thederivativeofx^{\{}-1/2\}is-\frac{1}{2}x^{\{}-3/2\}, andthederivativeof2xis2. Forthesecondpart, thederivativeo$
$x^{\{}-1\}isx^{\{}-2\}. Afterapplyingtheproductrule, wefindh'(x) = (x^{\{}-1/2\}+2x)(x^{\{}-$
$2\})+(7-x^{\{}-1\})(-\frac{1}{2}x^{\{}-3/2\}+2). Evaluatingthisatx = 4yieldsh'(4) = 12.987. Thisvalueindicatesthe$
4.
Note:$Theanswercorrectlyappliestheproductruleandchainruletodifferentiatethefunctionh(x)ande$
$4torevealtherateofchangeatthatpoint, whichisessentialforunderstandingthefunction'sbehavior.$

End of Answer Key